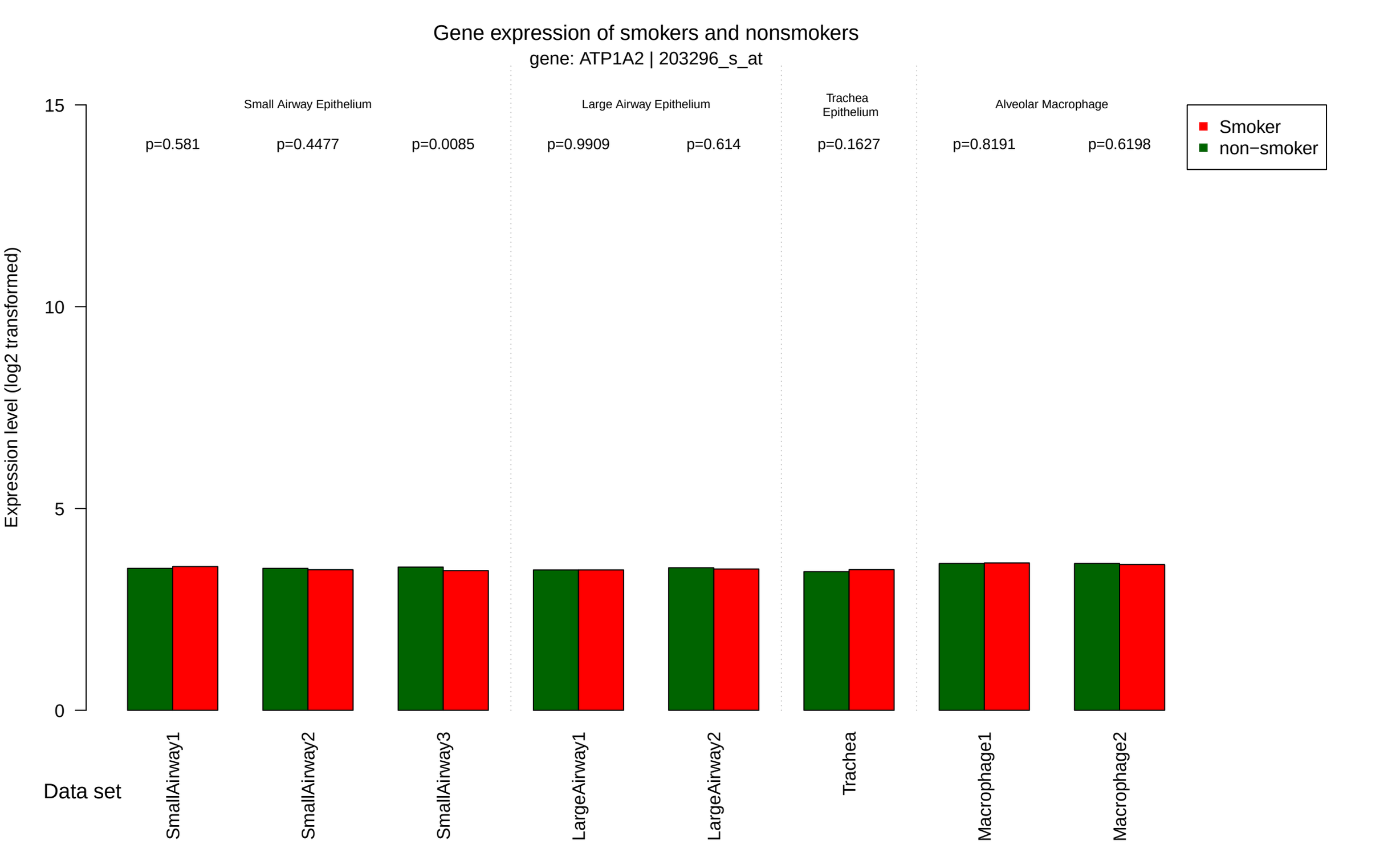
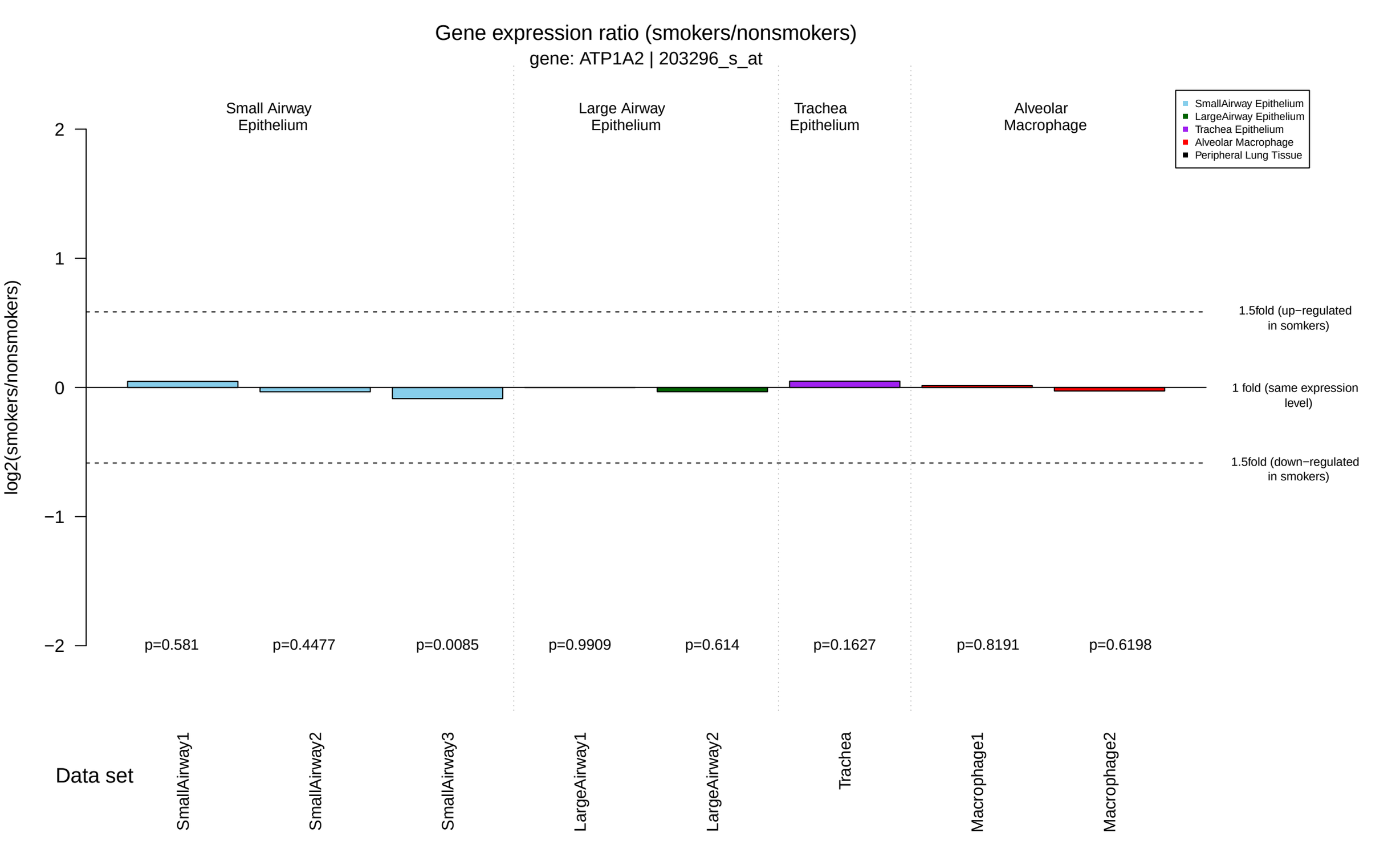
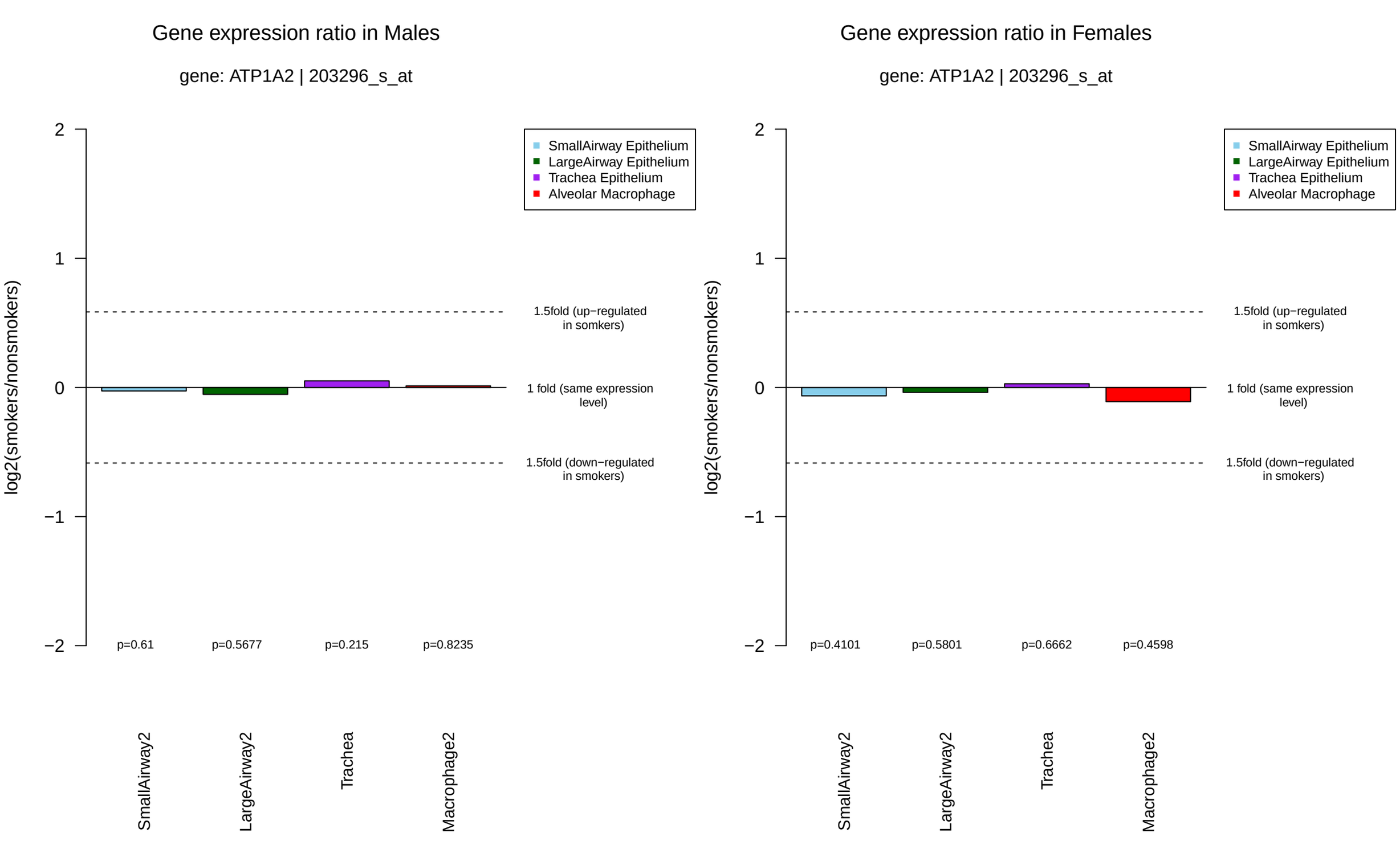
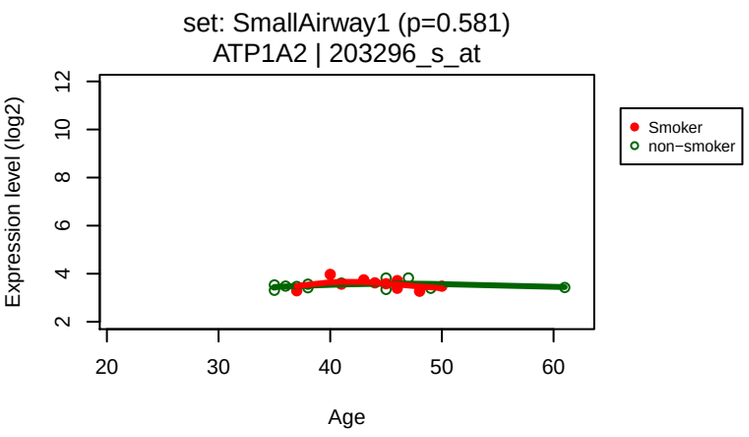
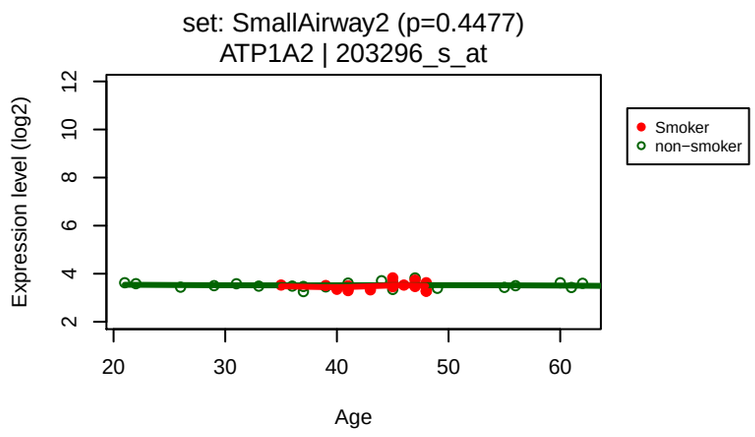
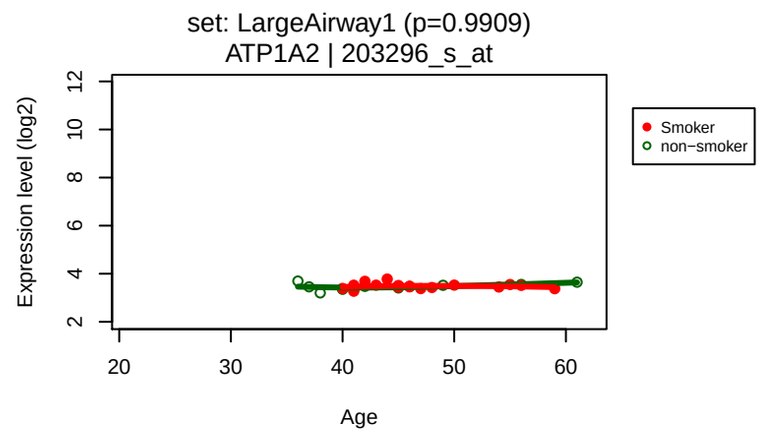

Gene expression of smokers and nonsmokers
gene: ATP1A2 | 203296_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.581
p=0.4477
p=0.0085
p=0.9909
p=0.614
p=0.1627
p=0.8191
p=0.6198
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

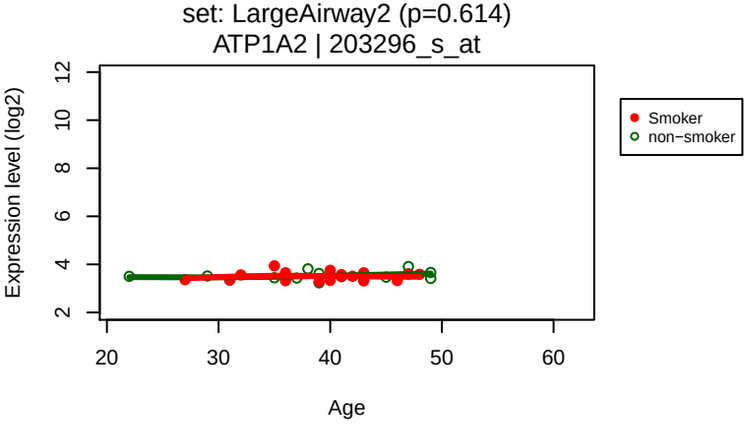

Gene expression ratio (smokers/nonsmokers)
gene: ATP1A2 | 203296_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.581
p=0.4477
p=0.0085
p=0.9909
p=0.614
p=0.1627
p=0.8191
p=0.6198
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

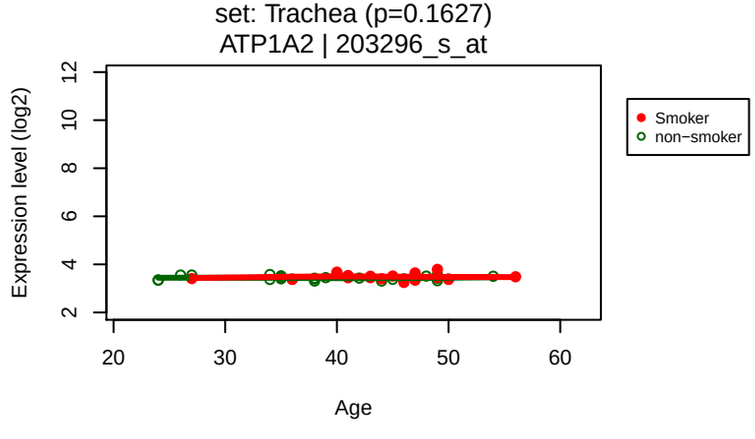

Gene expression ratio in Males
gene: ATP1A2 | 203296_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.61
p=0.5677
p=0.215
p=0.8235
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP1A2 | 203296_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4101
p=0.5801
p=0.6662
p=0.4598
SmallAirway2
LargeAirway2
Trachea
Macrophage2

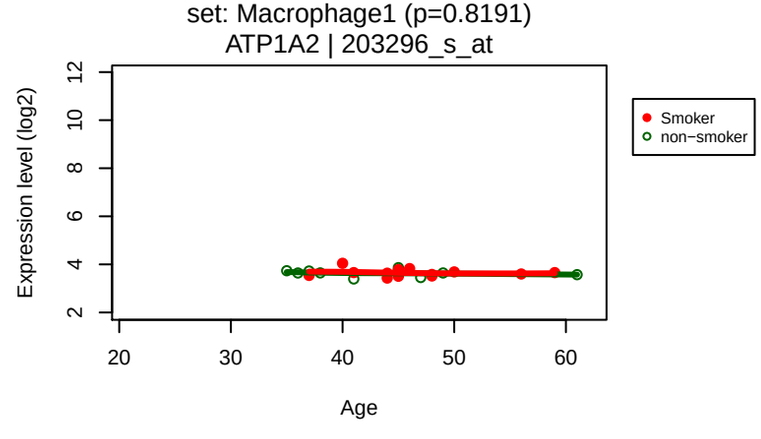

set: SmallAirway1 (p=0.581)
ATP1A2 | 203296_s_at
Expression level (log2)
Age
Smoker
non−smoker

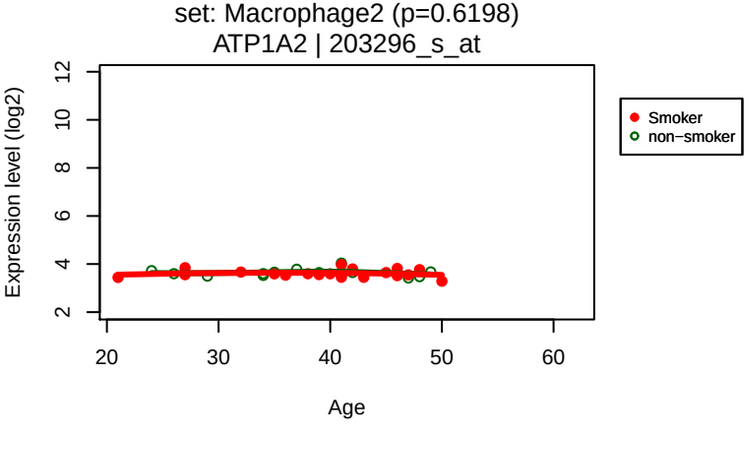

set: SmallAirway2 (p=0.4477)
ATP1A2 | 203296_s_at
Expression level (log2)
Age
Smoker
non−smoker

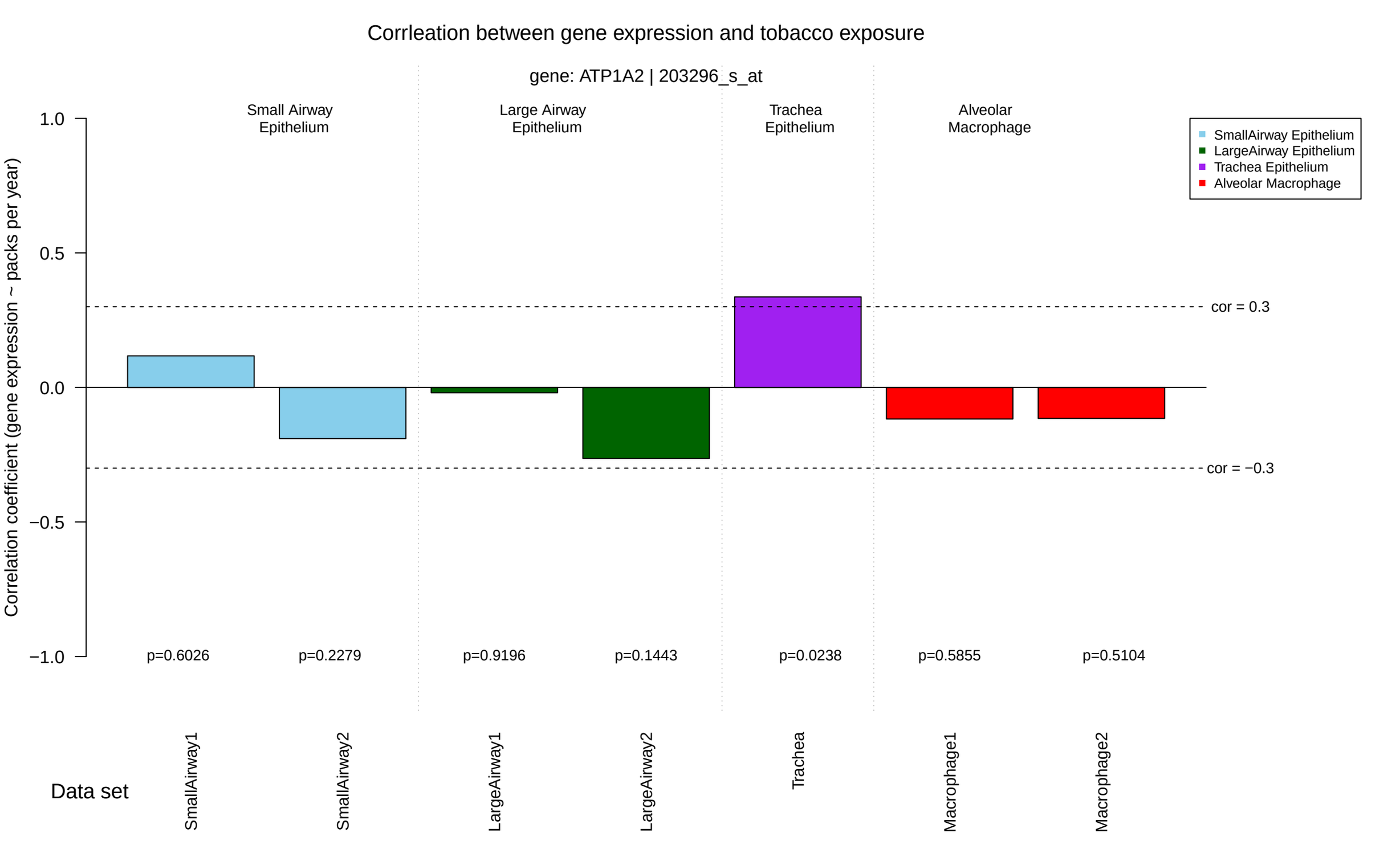

set: LargeAirway1 (p=0.9909)
ATP1A2 | 203296_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.614)
ATP1A2 | 203296_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1627)
ATP1A2 | 203296_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8191)
ATP1A2 | 203296_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6198)
ATP1A2 | 203296_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP1A2 | 203296_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6026
p=0.2279
p=0.9196
p=0.1443
p=0.0238
p=0.5855
p=0.5104
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2